1 Generator of Translations from the Unitary Operator

Spacetime translations are implemented by a unitary operator

$$U(a) = e^{-ia_\mu P^\mu}. \tag{1}$$

By definition, momentum eigenstates transform as

$$U(a)\,|p\rangle = e^{-ip\cdot a}|p\rangle. \tag{2}$$

Single-particle states are created from the vacuum via

$$|p\rangle = a^\dagger(p)\,|0\rangle, \tag{3}$$

and the vacuum is invariant under translations:

$$U(a)\,|0\rangle = |0\rangle. \tag{4}$$

Applying $U(a)$ to the state $|p\rangle$, we obtain

$$\begin{aligned} U(a)\,|p\rangle &= U(a)\,a^\dagger(p)\,|0\rangle && (5)\\ &= \big(U(a)\,a^\dagger(p)\,U^\dagger(a)\big)\,|0\rangle. && (6) \end{aligned}$$

Comparing with

$$U(a)\,|p\rangle = e^{-ip\cdot a}a^\dagger(p)\,|0\rangle, \tag{7}$$

we conclude that

$$U(a)\,a^\dagger(p)\,U^\dagger(a) = e^{-ip\cdot a}\,a^\dagger(p). \tag{8}$$

Similarly,

$$U(a)\,b^\dagger(p)\,U^\dagger(a) = e^{-ip\cdot a}\,b^\dagger(p). \tag{9}$$

Thus, $a^\dagger(p)$ and $b^\dagger(p)$ create momentum eigenstates with eigenvalue p^μ.

The operator that generates these transformations must therefore assign momentum p^μ to each particle. This is achieved by the number operators

$$N_a(p) = a^\dagger(p)a(p), \qquad N_b(p) = b^\dagger(p)b(p), \tag{10}$$

which count particles with momentum p.

Summing over all momenta, the total momentum operator is

$$P^\mu = \int d\tilde{p}\,p^\mu\left(a^\dagger(p)a(p) + b^\dagger(p)b(p)\right), \tag{11}$$

which correctly reproduces the transformation of all multiparticle states under translations.